

Lift Plan

Scaffold Module Installation

Multiple Hoist Operation

AS 2550.1-2011 section 6.27.4



NORTH JOHNSON RAIL BRIDGE – SPAN2

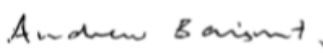
GGC Industries



Document Number | **4551 – Lift Plan – Span 2 – Scaffold Module Installation**

Revision | **01**

Rev	Date	By	Description of Changes
00	19/03/2026	JA	First Issue
01	21/03/2026	JA	Amended drawings and pick up points

	Name	Title	Signed	Date
Prepared by	Jerelle Aggaboa	Design Engineer		21/03/26
Reviewed by	Clinton Heane	Director		21/03/26
Certified by	Dr. Andrew Baigent RPEQ: 9984	Engineer		21/03/26

Related Documents

- 4551 – GGC – NJB Bridge - SWMS – REV05
- 4551 – Methodology – Span 2 – REV01
- 4551.5_REV01 – North Johnson Lifting Module Design
- 4551.2_REV05 - North Johnson Rail Bridge Span 2 drawings
- GGC-PLN-001 Health & Safety Management Plan
- Relevant client/site emergency procedures

Rigging Bill of Materials

Quantity	Description	Weight / unit	weight	Weight below hoist (to be lifted)	SWL
4	3.2 Tonne bow shackles	1kg	4kg	4kg	3.2 t
4	4.7 Tonne bow shackles	2kg	8kg		4.7 t
4	2 t x 2m soft sling	1kg	4kg	4kg	4.0 t
4	Kito 2 Tonne electric chain hoists 10m chain	54 kg	216 kg		2.0 t
4	2m x 8mm 80 grade chain snorter with master links	10kg	40 kg		4.0 T
4	10m x 10mm 80 grade chain below hoist	21kg	81kg	81kg	2 T
4	2 Tonne Hook assembly	2.75kg	11kg	11kg	2T
	Total Below Hoist Weight (weight to be lifted)			100kg	

Weight of loads to be lifted	
Section 1	3.502 tonne
Section 2	3.117 tonne
Weight of rigging gear below the hoist	100kg
Weight of 2 x scaffolder	200kg
Maximum weight to be lifted	3.802 Tonne
Maximum weight per hoist	0.951Tonne
Maximum capacity of weakest part of rigging	47.55%

Hoist 1	Kito electric chain hoist
Serial Number	ERIA-630 234688
Capacity	2 Tonne
Winch speed	Single speed -3.7m / minute
Chain details	10m x 10mm 80 Grade
Weight	73kg
Supply	415 Volt - 3 phase - 50 Cycle

Hoist 2	Kito electric chain hoist
Serial Number	ERIA-630 269497
Capacity	2 Tonne
Winch speed	Single speed - 3.7m / minute
Chain details	10m x 10mm 80 Grade
Weight	73kg
Supply	415 Volt - 3 phase - 50 Cycle

Hoist 3	Kito electric chain hoist
Serial Number	ERIA-630 269525
Capacity	2 Tonne
Winch speed	Single speed -3.7m / minute
Chain details	10m x 10mm 80 Grade
Weight	73kg
Supply	415 Volt - 3 phase- 50 Cycle

Hoist 4	Kito electric chain hoist
Serial Number	ERIA-630 234688
Capacity	2 Tonne
Winch speed	Single speed -3.7m / minute
Chain details	10m x 10mm 80 Grade
Weight	73kg
Supply	415 Volt - 3 phase - 50 Cycle

Lift configuration	4 point synchronized lift
Angle of lifting chain	~10 Degrees – 1.54% increase in tension
Maximum total weight to be lifted	3.802 Tonne
Maximum weight per hoist	0.951
Increase in capacity due to angle	0.966
Max capacity of winch	48.3%
Maximum capacity of weakest link	48.3%
Maximum point load on support beams for lift	0.951 Tonne

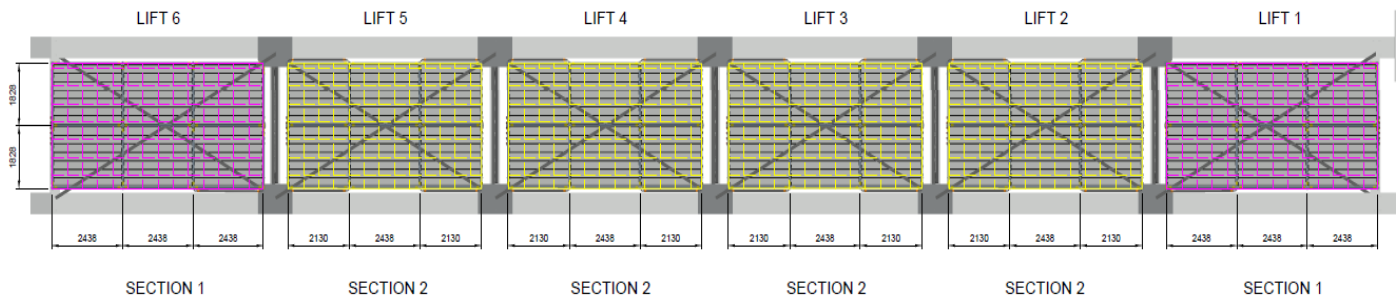
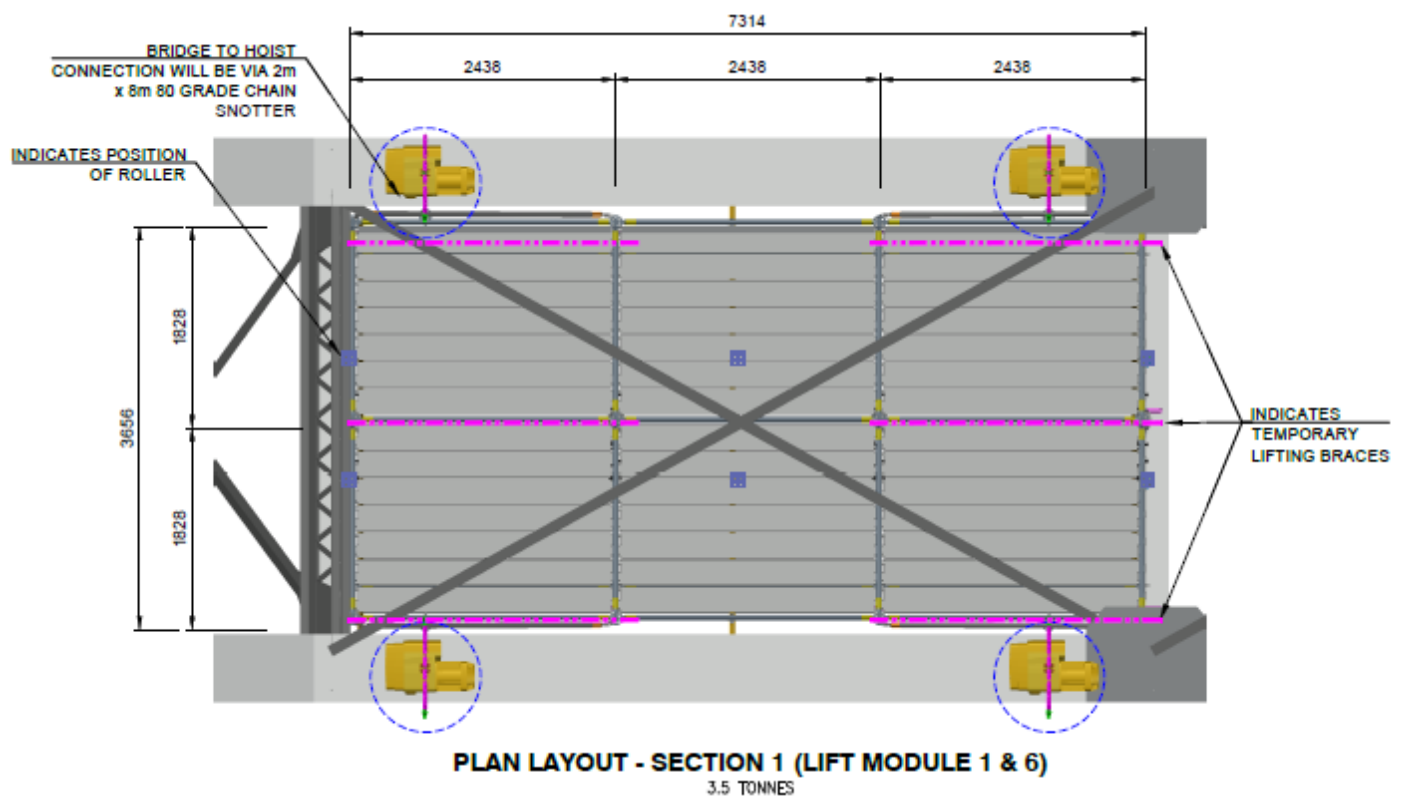


Figure 1. 6 Modules to be lifted in Position

Lifting section type 1:

- 3.7m x 7.3m
- 2 Lifting Modules (Refer to Figure 1 – Lift 1 & 6)

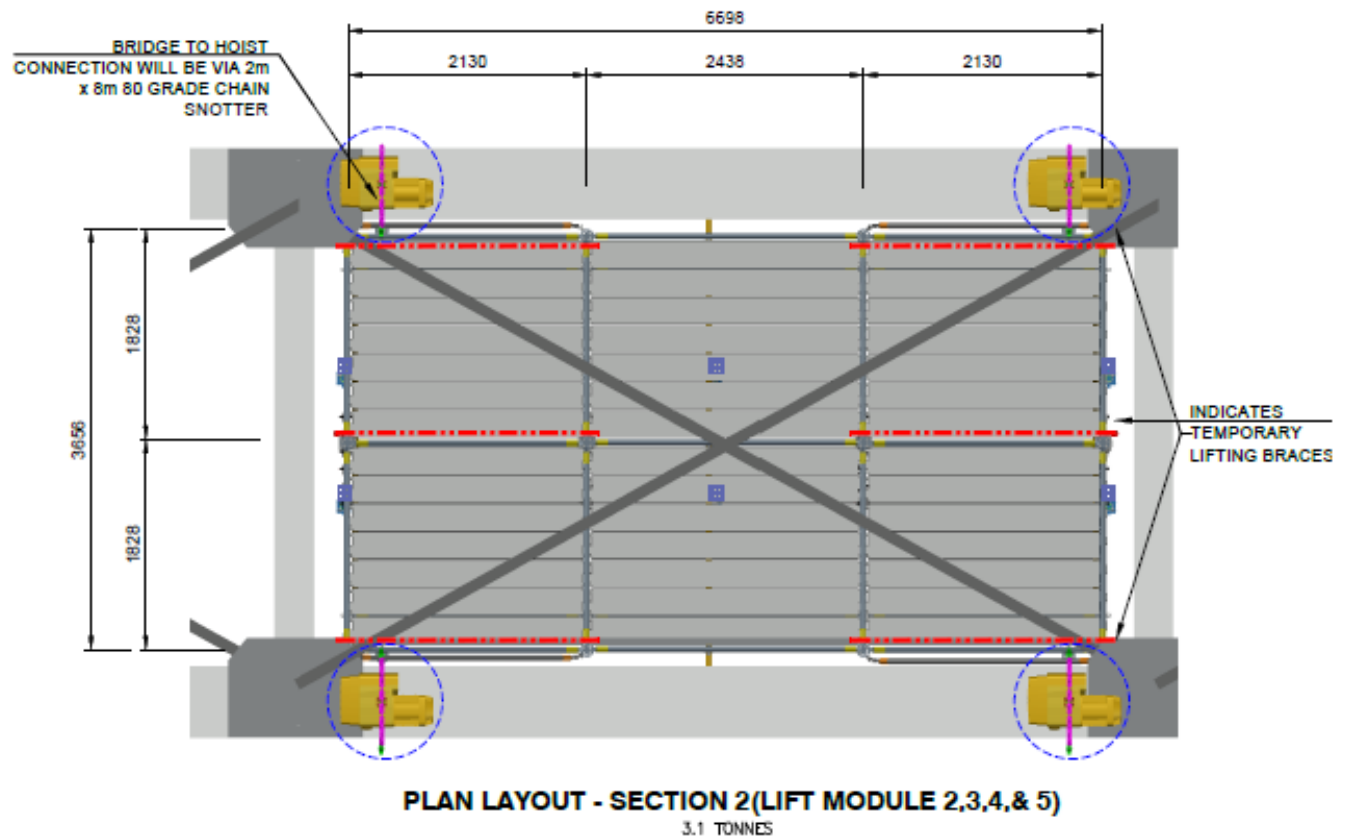


Lifting Section type 1: Bill Of Materials

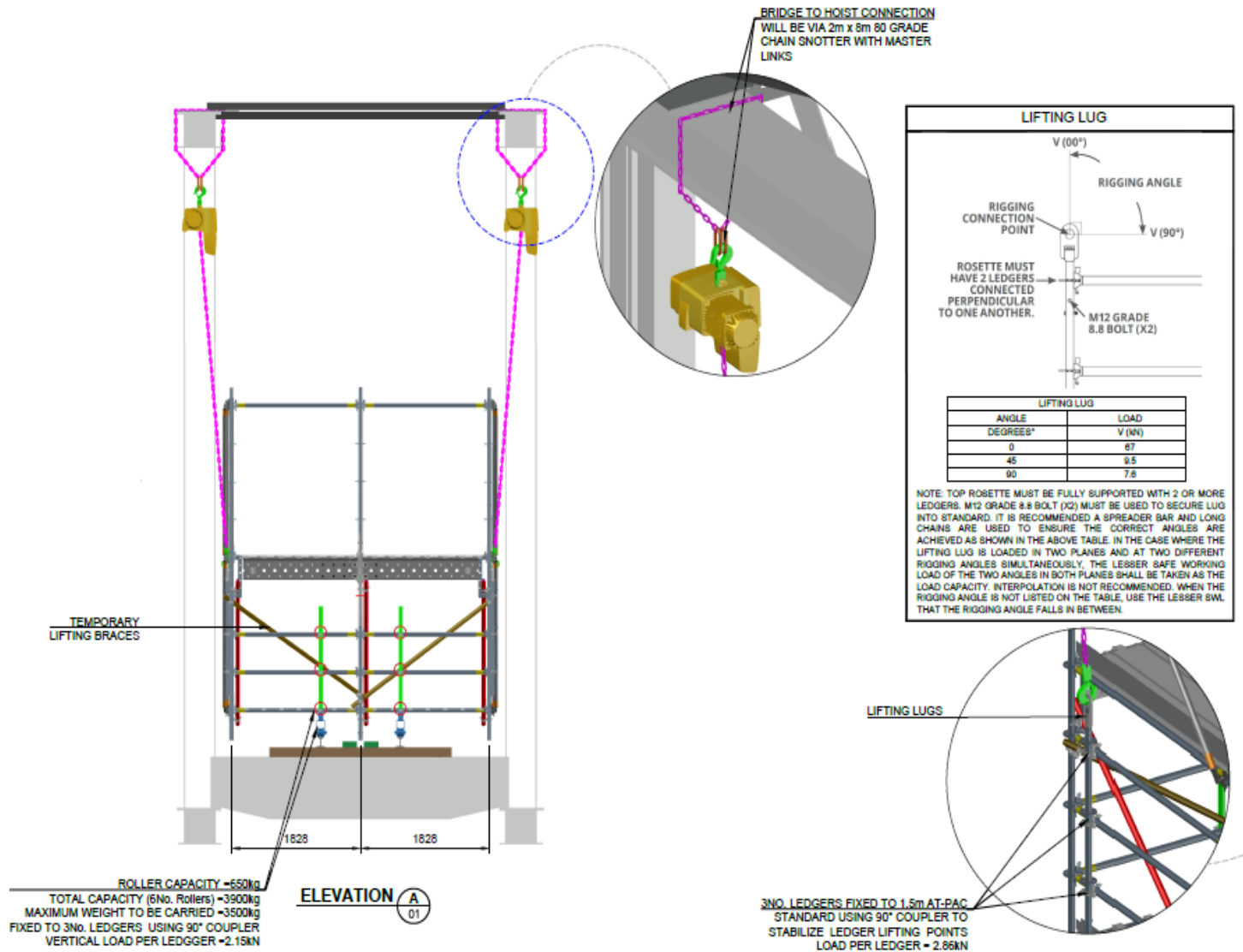
Part Number	Description	Quantity	Weight (ea)	Total Weight
01.01.150.00	Standard (3 Ring) With Crimped Spigot	16	7.94	127.04
01.01.200.00	Standard (4 Ring) With Crimped Spigot	4	10.32	41.28
01.01.300.00	Standard (6 Ring) With Crimped Spigot	8	14.99	119.92
01.03.182.00	Ledgers -Ledger O-Type 1.82m	26	7.25	188.5
01.03.243.00	Ledgers -Ledger O-Type 2.43m	68	9.52	647.36
01.06.243.00	Fixed Braces - Bay Brace 2.0m x 2.43m	4	10.14	40.56
08.02.243.00	Steel Planks -Steel Plank 240mm - O-Type 2.43m	84	15.98	1342.32
08.07.243.00	Infill Plank - 2.43m	18	10.43	187.74
08.08.243.00	Interlocking Toe board 2.43m	12	7.19	86.28
60.01.050.00	MPBS Alum Beam 1.52m/5'	4	19.13	76.52
60.01.060.00	MPBS Alum Beam 1.83m/6'	4	22.99	91.96
60.02.002.00	MPBS Beam Splice (Aluminium)	4	10.62	42.48
60.07.001.00	MPBS End Beam Adaptor (Ring lock)	8	6.33	50.64
60.06.001.00	MPBS End Beam Adaptor (Adjustable)	8	6.29	50.32
60.09.002.00	MPBS Top Standard Connector	4	8.47	33.88
60.10.001.00	MPBS Bottom Standard Connector	4	7.10	28.4
60.17.001.00	MPBS Ledger Connector	12	.72	8.64
60.04.001.00	MPBS Pin (125mm)	72	.29	20.88
60.04.002.00	MPBS Safety Clip	72	.03	2.16
KSST15	Tube 1.5 meter	6	12.00	72
KSST30	Tube 3.3 meter	6	13.60	81.6
KSST24	Tube 2.4 meter	4	9.60	38.4
KSST42	Tube 4.2 meter	3	16.80	50.4
TCSWB22	Swivel Coupler	20	1.60	32
TCRAB22	90 Degree Coupler	27	1.60	32
	Rollers	6	1.50	9
		507		3502.28

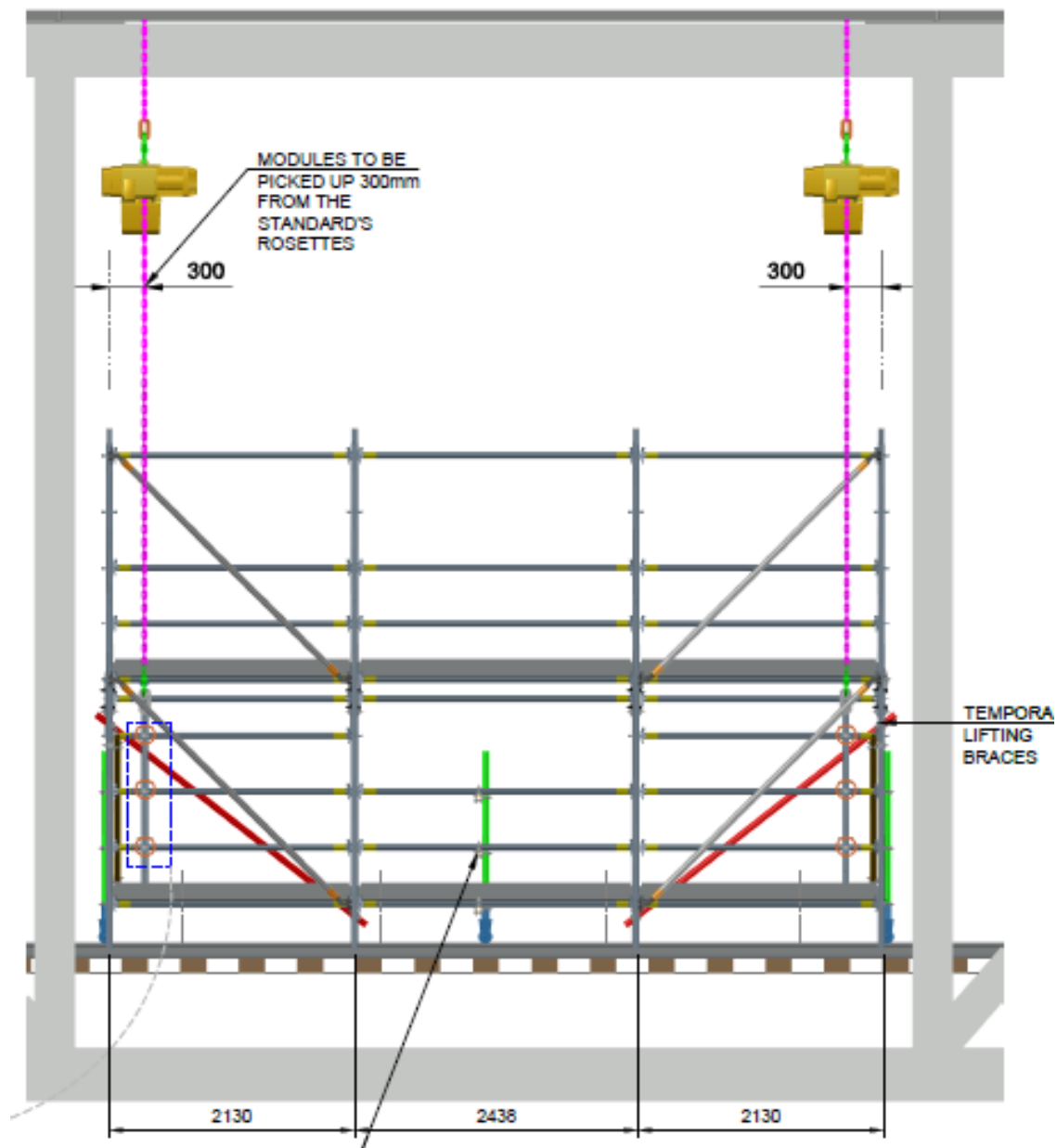
Lifting section type 2:

- 3.7m x 6.7m
- 4 Lifting Modules (Refer to Figure 1 – Lift 2, 3, 4, 5)



Elevation 1:



Elevation 2:

Lifting Section type 2: Bill of materials

Part Number	Description	Quantity	Weight (ea)	Total Weight
01.01.150.00	Standard (3 Ring) With Crimped Spigot	16	7.94	127.04
01.01.200.00	Standard (4 Ring) With Crimped Spigot	4	10.32	41.28
01.01.300.00	Standard (6 Ring) With Crimped Spigot	8	14.99	119.92
01.03.182.00	Ledgers -Ledger O-Type 1.82m	26	7.25	116
01.03.213.00	Ledgers -Ledger O-Type 2.13m	38	8.40	285.6
01.03.243.00	Ledgers -Ledger O-Type 2.43m	40	9.52	161.84
01.06.213.00	Bay Brace 2.0m x 2.13m	8	9.52	76.16
08.02.213.00	Steel Plank 240mm - O-Type 2.13m	56	14.33	802.48
08.02.243.00	Steel Plank 240mm - O-Type 2.43m	28	15.98	447.44
08.07.213.00	Infill Plank - 2.13m	12	9.19	110.28
08.07.243.00	Infill Plank - 2.43m	6	10.43	62.58
08.08.213.00	Interlocking Toe board 2.13m	6	6.33	37.98
08.08.243.00	Interlocking Toe board 2.43m	3	7.19	21.57
60.01.050.00	MPBS Alum Beam 1.52m/5'	4	19.13	76.52
60.01.060.00	MPBS Alum Beam 1.83m/6'	4	22.99	91.96
60.02.002.00	MPBS Beam Splice (Aluminium)	4	10.62	42.48
60.07.001.00	MPBS End Beam Adaptor (Ring lock)	8	6.33	50.64
60.06.001.00	MPBS End Beam Adaptor (Adjustable)	8	6.29	50.32
60.09.002.00	MPBS Top Standard Connector	4	8.47	33.88
60.10.001.00	MPBS Bottom Standard Connector	4	7.10	28.4
60.17.001.00	MPBS Ledger Connector	12	.72	8.64
60.04.001.00	MPBS Pin (125mm)	72	.29	20.88
60.04.002.00	MPBS Safety Clip	72	.03	2.16
KSST15	Tube 1.5 meter	6	12.00	72
KSST30	Tube 3.0 meter	6	12.00	72
KSST24	Tube 2.4 meter	4	9.60	38.4
TCSWB22	Swivel Coupler	20	1.60	32
KSST42	Tube 4.2 meter	3	16.80	50.4
TCSWB22	Swivel Coupler	20	1.60	32
TCRAB22	90 Degree Coupler	27	1.60	32
	Rollers	6	1.50	9
		443		3,117.85

CALCULATIONS

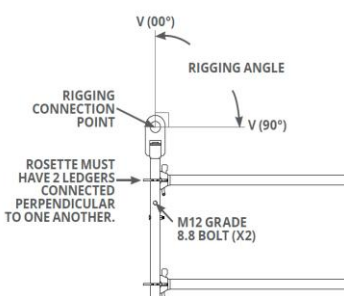
A. Capacities of Roller

Weight of loads to be carried by Roller	
Section 1	3.502 tonne
Section 2	3.117 tonne
Capacity of Roller	750kg
Capacity of 6 Rollers	4500 kg ~ 4.5 tonne
Conclusion	The rollers have adequate capacity to sustain the applied loads.

B. Ledgers

Load Transferred to Ledgers																																														
Section 1	34.35kN																																													
Section 2	30.58kN																																													
Capacity of Ledgers	0 LEDGERS/TRANSOMS <table><tr><th colspan="3">0 LEDGERS/TRANSOMS</th></tr><tr><th>LENGTH</th><th>POINT LOAD</th><th>UDL</th></tr><tr><th>(m)</th><th>V (kN)</th><th>w (kN/m)</th></tr><tr><td>0.35</td><td>15.0</td><td>30.00</td></tr><tr><td>0.60</td><td>13.70</td><td>29.25</td></tr><tr><td>0.65</td><td>13.26</td><td>29.00</td></tr><tr><td>0.84</td><td>10.32</td><td>24.30</td></tr><tr><td>0.88</td><td>9.70</td><td>23.31</td></tr><tr><td>1.06</td><td>8.59</td><td>18.85</td></tr><tr><td>1.15</td><td>7.92</td><td>17.26</td></tr><tr><td>1.57</td><td>6.00 *</td><td>9.73 *</td></tr><tr><td>1.82</td><td>5.17 *</td><td>7.44 *</td></tr><tr><td>2.13</td><td>4.37 *</td><td>5.59 *</td></tr><tr><td>2.43</td><td>3.97 *</td><td>3.70 *</td></tr><tr><td>3.05</td><td>3.22 *</td><td>2.26 *</td></tr></table>	0 LEDGERS/TRANSOMS			LENGTH	POINT LOAD	UDL	(m)	V (kN)	w (kN/m)	0.35	15.0	30.00	0.60	13.70	29.25	0.65	13.26	29.00	0.84	10.32	24.30	0.88	9.70	23.31	1.06	8.59	18.85	1.15	7.92	17.26	1.57	6.00 *	9.73 *	1.82	5.17 *	7.44 *	2.13	4.37 *	5.59 *	2.43	3.97 *	3.70 *	3.05	3.22 *	2.26 *
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	Reference: AT- Pac Technical Manual																																													
Capacity of 18No. 1.82m Ledgers	5.17kN x 6 = 93.06 kN																																													
Conclusion	The ledgers have adequate capacity to sustain the applied loads.																																													

C. Lifting Lugs

Load Carried by Lifting Lugs													
Section 1	34.35kN / 4 Locations of Lifting Lugs = 8.59kN												
Section 2	30.58kN/ 4 Locations of Lifting Lugs = 7.65kN												
Capacity of Lifting Lugs	LIFTING LUG <table border="1"> <thead> <tr> <th colspan="2">LIFTING LUG</th> </tr> <tr> <th>ANGLE</th><th>LOAD</th> </tr> <tr> <th>Degrees</th><th>V (kN)</th> </tr> </thead> <tbody> <tr> <td>0</td><td>67</td> </tr> <tr> <td>45</td><td>9.5</td> </tr> <tr> <td>90</td><td>7.6</td> </tr> </tbody> </table> <i>Note: Top Rosette must be fully supported with 2 or more Ledgers. M12 Grade 8.8 Bolt (x2) must be used to secure Lug into Standard. It is recommended a spreader bar and long chains are used to ensure the correct angles are achieved as shown in the above table.</i> <i>In the case where the lifting lug is loaded in two planes and at two different rigging angles simultaneously, the lesser Safe Working Load of the two angles in both planes shall be taken as the load capacity.</i> <i>Interpolation is not recommended. When the rigging angle is not listed on the table, use the lesser SWL that the rigging angle falls in between.</i> <i>Test Report: SHIN170200375CCM (F.O.S. 1.5:1)</i>  Reference: AT- Pac Technical Manual	LIFTING LUG		ANGLE	LOAD	Degrees	V (kN)	0	67	45	9.5	90	7.6
LIFTING LUG													
ANGLE	LOAD												
Degrees	V (kN)												
0	67												
45	9.5												
90	7.6												
Capacity of Lifting Lugs @ 0 deg Rigging Angle	67kN												
Conclusion	The lifting lugs have adequate capacity to sustain the applied loads.												

D. Coupler Slip Capacity

Load Carried by Lifting Lugs	
Section 1	34.35kN / 4 Locations of Lifting Lugs = 8.59kN
Section 2	30.58kN/ 4 Locations of Lifting Lugs = 7.65kN
Slip Capacity of Couplers	6kN
Slip Capacity of 3No. Couplers	18kN
Conclusion	The couplers have adequate capacity to sustain the applied loads.

Remote radio control data:

- Juunko K808 radio remote handset Serial # C2EQ
- HS – series receiver
- 20 KVA Genset – TBC

Review and Sign-Off

By signing the below, you acknowledge you have read and understand the lift plan

Name	Qualifications	Signature	Date